

XD12

The LED Current Sink Driver IC

12 Channels

General Description

The XD12 is a 12-channel constant current sink drivers for mini-LED Backlight display applications in AM direct driving methodology. It supports max 64mA and 40V at each channel output pin rating. This device can support max 14-bit PWM dimming resolution and a 12-bit DAC global brightness resolution.

The self-protections features, LED open/short detection and thermal shutdown, are embedded. In addition, a Feedback control for adjusting a VLED power is included.

It is used with a XC24 controller as a companion chip. A lot of the XD12 can be daisy to a XC24 channels and controlled through a serial interface by a XC24 controller.

Application

LED back-lighting with local dimming

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Features

- Bi-directional 1-wire interface and configurable transmit rate
- Daisied connection between devices and pin layout suitable for single layer PCB
- 12 independent channels in a package
- Maximum 40V headroom voltage
- Configurable max current level (4/8/12/16/24/32/46/64mA)
- 12-bit DAC global brightness control
- 14-bit PWM dimming
- Configurable dimming directions, Head shift lighting and Tail shift lighting
- Sync controllability between daisied devices
- Sink current accuracy $\pm 2\%$
- Configurable LED short detection level and open detection
- Configurable feedback generating level
- Configurable over-temperature shutdown @ Typical 135°C
- Supply voltage 4.5 ~ 5.5V
- 24-PIN QFN 3030 package
- Latch-up Immunity, I_{LATCH} (JEDEC JESD78D Nov 2011) : $\pm 100\text{mA}$
- ESD protection, ESD_{HBM} (Norm: MIL 883 E Method 3015 Human Body Model) : $\pm 2\text{kV}$

Block Diagram

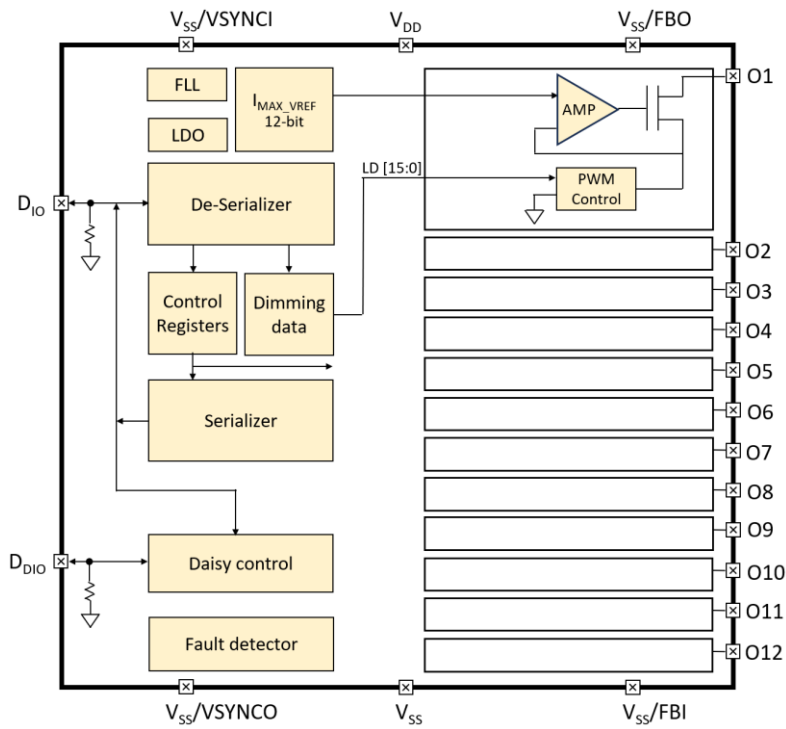


Figure 1. Block Diagram

Application Circuit

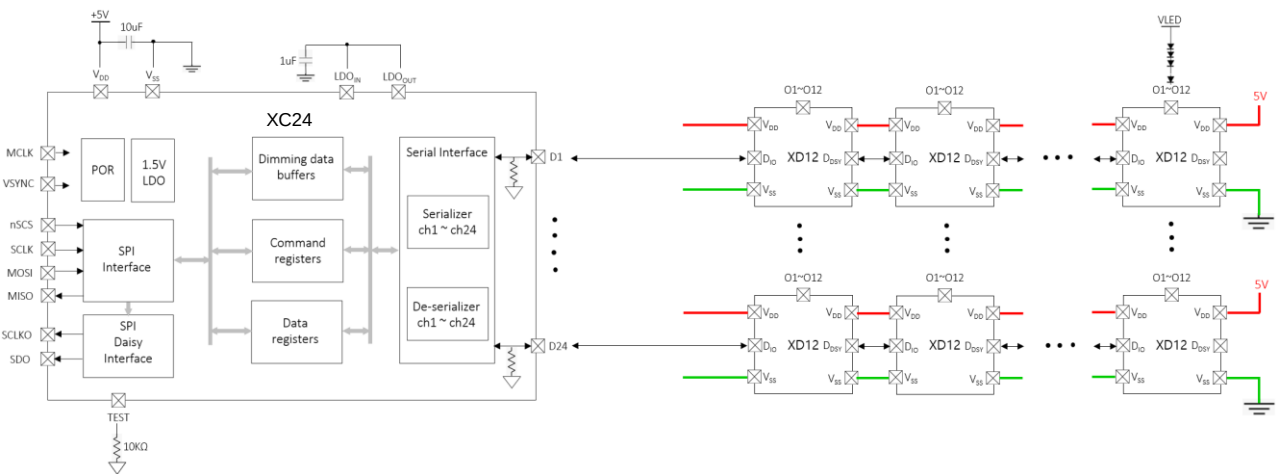


Figure 2. Application scheme

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Pin diagram (Top View)

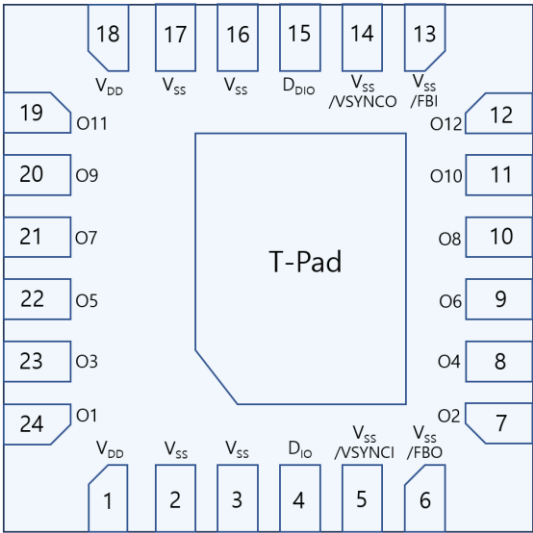


Figure 3. Pin diagram

Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
V _{DD}	1	-	Power	Supply voltage	
V _{SS}	2	-	GND	Ground	
V _{SS}	3	-	GND	Ground	
D _{IO}	4	I/O	Digital	Data In/Out	Internal PD, 48KΩ
V _{SS} /VSYNCI ¹⁾	5	I	GND/ Digital	Ground/ VSYNCI	Internal PD, 48KΩ
V _{SS} /FBO ¹⁾	6	O	GND/ Digital	Ground/ FBO	Open-drain output
O2	7	O	Current Load	Current Sink 2	If unused, connect to GND
O4	8	O	Current Load	Current Sink 4	If unused, connect to GND
O6	9	O	Current Load	Current Sink 6	If unused, connect to GND
O8	10	O	Current Load	Current Sink 8	If unused, connect to GND
O10	11	O	Current Load	Current Sink 10	If unused, connect to GND
O12	12	O	Current Load	Current Sink 12	If unused, connect to GND
V _{SS} /FBI ¹⁾	13	I	GND/ Digital	Ground/ FBI	Internal PU, 48KΩ
V _{SS} /VSYNCO ¹⁾	14	O	GND/Digital	Ground/ VSYNCO	

Table 1. Pin Description

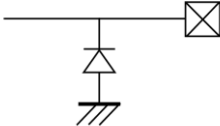
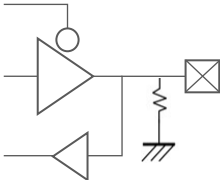
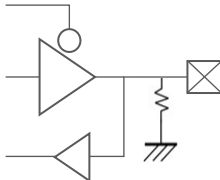
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Pin description

Pin Name	Pin No.	I/O	Type	Description	Remark
D _{DIO}	15	I/O	Digital	Daisied Data In/Out	Internal PD, 48KΩ
V _{SS}	16	-	GND	Ground	
V _{SS}	17	-	GND	Ground	
V _{DD}	18	-	Power	Supply Voltage	
O11	19	O	Current Load	Current Sink 11	If unused, connect to GND
O9	20	O	Current Load	Current Sink 9	If unused, connect to GND
O7	21	O	Current Load	Current Sink 7	If unused, connect to GND
O5	22	O	Current Load	Current Sink 5	If unused, connect to GND
O3	23	O	Current Load	Current Sink 3	If unused, connect to GND
O1	24	O	Current Load	Current Sink 1	If unused, connect to GND

Note : 1) In case of using FB and VSYNC function, it needs to set the FBO_E and EXT_VS_E bit. If not using, it is recommended to connect to GND. The FBO pin is an open-drain port.

Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
V _{DD}	-		Supply voltage
D _{IO}	I/O		Data In/Out
D _{DIO}	I/O		Daisied Data In/Out

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Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit	Description
O1 ~ O12	Current Sink		Current Load
VSS	-		Ground
VSS/ VSYNCI	Input		Ground or VSYNCI
VSS/ VSYNCO	Output		Ground or VSYNCO
VSS/ FBI	Input		Ground or FBI
VSS/ FBO	Output		Ground or FBO

Figure. 4 Equivalent Circuits of individual pins

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Maximum Absolute ratings (Ta = 25°C unless otherwise noted)

Parameter	Symbol	Value	Unit
Supply voltage	V _{DD}	-0.3 ~ 6.5	V
Sink output pin voltage	V _O	40	V
Sink output current	I _O	64	mA
D _{IO} to GND	V _{DIO}	-0.3 ~ V _{DD} +0.3	V
D _{DIO} to GND	V _{DDIO}	-0.3 ~ V _{DD} +0.3	V

Thermal Resistance

Junction to Ambient	R _{θJA}	TBD ¹⁾	°C/W
Junction to Case	R _{θJC}	TBD ¹⁾	°C/W
Power Dissipation on package	P _{TOT}	TBD ¹⁾	W
Junction temperature	T _J	-20 ~ 150	°C
Storage temperature	T _{STG}	-45 ~ 150	°C
Relative Humidity (non-condensing)	RH _{NC}	5 ~ 85	%
Moisture Sensitivity	MSL	Maximum Floor Life-time	3

Table 2. Maximum Absolute ratings

Stresses beyond the Maximum Absolute ratings may cause permanent damage to the device
Note : 1) Surface mounted on FR-4 substrate PC Board 1" x 1" inch 2-oz copper plate

Electrostatic Discharge

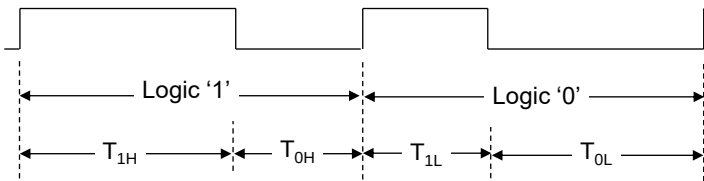
Parameter	Symbol	Value	Unit	Remark
Electrostatic Discharge HBM ; Contact	ESD _{HBM}	±2K	V	C=100 pF, R=1.5 kΩ
Electrostatic Discharge CDM ; Contact	ESD _{CDM}	±500	V	R=1 Ω

Table 3. Electrostatic Discharge

Electrical Characteristic (V_{DD} =5.0V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Supply voltage						
Operating voltage	V _{DD} ¹⁾		4.5	5.0	5.5	V
Supply current						
Quiescent Current	I _s	Standby state, No LED current	-	-	1.7	mA
Operating Current 1	I _{DD1}	All output sink current = 4mA	-	-	2.5	mA
Operating Current 2	I _{DD2}	All output sink current = 8mA	-	-	2.8	mA
Operating Current 3	I _{DD3}	All output sink current = 12mA	-	-	3.1	mA
Operating Current 4	I _{DD4}	All output sink current = 16mA	-	-	3.3	mA
Operating Current 5	I _{DD5}	All output sink current = 24mA	-	-	3.8	mA
Operating Current 6	I _{DD6}	All output sink current = 32mA	-	-	4.3	mA
Operating Current 7	I _{DD7}	All output sink current = 46mA	-	-	5.2	mA
Operating Current 8	I _{DD8}	All output sink current = 64mA	-	-	6.4	mA
Oscillator						
Internal MCLK	f _{OSC}			40		MHz
In/Out						
D _{IO} , D _{DIO}	V _{IL} V _{IH}	Input logic low level	-	-	0.5	V
		Input logic high level	V _{DD} * 0.8	-	V _{DD}	
	V _{OL} V _{OH}	Output logic low level	-	-	0.5	V
		Output logic high level	V _{DD} * 0.8	-	V _{DD}	
		R _{PD}	Internal pull-down resistance	-	48	-
Logic high time ²⁾	T _{1H} , T _{0L}		6	8	20	MCLK
Logic low time ²⁾	T _{1L} , T _{0H}		3	4	10	MCLK
Propagation delay	T _{D_DS}	Downstream (D _{IO} to D _{DIO})	-	-	5.0	ns
Propagation delay	T _{D_US}	Upstream (D _{DIO} to D _{IO})	-	-	5.5	ns

Table 4. Electrical Characteristic



Note : 1) The operation below V_{DD}, 5V will limit a maximum LED sinking current and affect a current variation.

2) T_{1H} ≥ T_{0H} * 2, T_{0L} ≥ T_{1L} * 2, (T_{1H} + T_{0H}) = (T_{0L} + T_{1L}) ≤ 31

Electrical Characteristic (V_{DD}=5.0V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
Output						
Driving voltage	V _O	Headroom voltage V _{HD} : 0.55V@ I _O 4mA, 0.60V@ I _O 8mA 0.65V@ I _O 12mA, 0.70V@ I _O 16mA 0.75V@ I _O 24mA, 0.80V@ I _O 32mA 0.85V@ I _O 46mA, 0.90V@ I _O 64mA	V _{HD}	-	40	V
Short detection voltage ³⁾	V _{SD}	Short detection level V _{SD} : 3b'000=3V, 3b'001=4V 3b'010=6V, 3b'011=8V 3b'100=12V, 3b'101=16V 3b'110=24V, 3b'111=36V	V _{SD}	-	-	V
Open detection voltage ³⁾	V _{OD}		-	-	0.25	V
Sink current	I _O	Device max current level : I _O 3'b000 = 4mA, 3'b001 = 8mA 3'b010 = 12mA, 3'b011 = 16mA 3'b100 = 24mA, 3'b101 = 32mA 3'b110 = 46mA, 3'b111 = 64mA	-	-	64	mA
Current accuracy ⁴⁾	%	@ I _O =Imax(8mA)*0.02	-2	-	+2	%
		@ I _O = Imax(8mA)*0.1	-2	-	+2	%
		@ I _O = Imax(8mA)*1	-2	-	+2	%
LED off current	I _{OFF}	V _O = 40V	-	-	300	nA
Feedback						
FB threshold voltage (Logically ORed for 12-ch)	V _{FBA}	FB level V _{FBA} : 3'b000 = 0.40V, 3'b001 = 0.50V 3'b010 = 0.60V, 3'b011 = 0.70V 3'b100 = 0.85V, 3'b101 = 1.00V 3'b110 = 1.15V, 3'b111 = 1.30V	-	V _{FBA}	-	V
FB hysteresis voltage	V _{FBH}		40	60	110	mV
Thermal protection						
Thermal cut off @T _J	T _{CUT}	OFS_TEMP [3:0] = 0x8	-	135	-	°C
Thermal cut off hysteresis @T _J	T _{HYS}		10	20	30	°C

Note : 3) If LED open or short event is happened, the corresponding channel will be turned-off under the function enabled. If the thermal is happened, all channels output will be turned-off under the function enabled.
4) Under satisfying the power dissipation specification.

Electrical Characteristic (V_{DD}=5.0V, Ta=25.0°C unless otherwise noted)

Characteristics	Symbol	Condition	Value			Unit
			min	typ	max	
V _{DD} Ramp up						
V _{DD} Ramp up time	T _{VDD}				0.3	ms
V _{DD} Sequence						
Internally Initial time	T _{INIT}		10			ms
Allowed 1 st command	T _{FCMD}	after the V _{DD} reaches 5V	10			ms
Configuration Setup time	T _{CFG_SU}	depends on the daisied devices #		500		ms
Off time	T _{OFF}	after the V _{LED} goes to 0	50			ms
Timeout : Internal timer starts after completing to receive a read-back command and stops after first response bit from a last daisied device						
Internal Timeout ⁵⁾	T _{TIMEOUT}			820		us
VSYNC daisy out @ EXT_VS_E = 1						
VSYNCO delay	T _{VSOD}	from VSYNCI to VSYNCO			4.5	ns

Note : 5) If the timeout will be expired, the internal communication logic will go to an initial state, so new command transaction will be possible. The timeout function is configurable.

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

Power Sequence

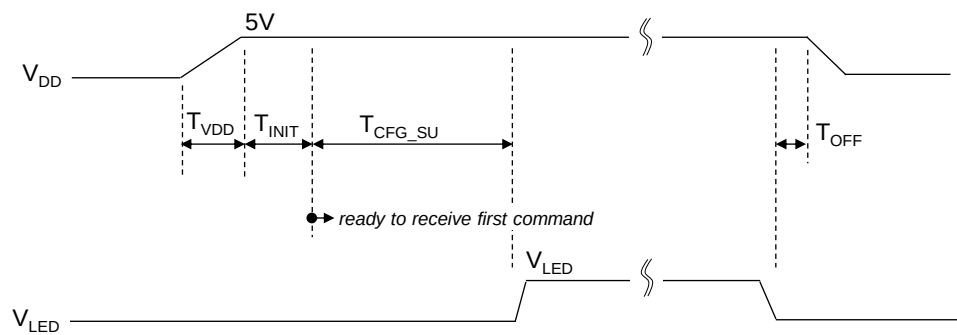


Figure. 5 Ramp-up and Power On/Off sequence

Functional Description

Serial Communication Protocol

The XD can be controlled with several commands through a XC controller or directly by a Micom. The access protocol and frame format are as follows and if several XD is connected in daisy, the commands should be repeated successively as many as the XD number in daisy chain.

PROTOCOL		frame format																					
write command	1101	bit21	bit20	bit19	bit18	bit17	bit16	bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
		start	cmd code [2:0]				addr [5:0]					data [11:0]											
read command	1110	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0												
		start	cmd code [2:0]				addr [5:0]																
readout response	1110	bit20	bit19	bit18	bit17	bit16	bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
		start	cmd code [2:0]				ld [4:0]					data [11:0]											
ld transmit	1111	bit19	bit18	bit17	bit16	bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0		
		start	cmd code [2:0]				ld [15:0]																
fault read command	1010	bit8	bit7	bit6	bit5																		
		start	cmd code [2:0]																				
fault readout response	1010	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0														
		start	cmd code [2:0]				thermal	short	open	fb													
sync gen command	1001	bit4	bit3	bit2	bit1																		
		start	cmd code [2:0]																				
id gen	1000	bit4	bit3	bit2	bit1																		
		start	cmd code [2:0]																				

Table 5. Serial communication protocol

PWM Configuration

The PWM behavior depends on the PWM configuration set to the PWM_RES and SCAN_NO [2:0] as shown in the Table 5.

☒ don't care ☐ valid bit

PWM_RES	SCAN_NO [2:0]	sub- frame #	LD Data [15:0]																step # /sub-frame	pwm freq hz@120
			D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0		
0 (12-bit)	0	1	x	x	x	x	11	10	9	8	7	6	5	4	3	2	1	0	4,095	120
	1	2	x	x	x	x	x	10	9	8	7	6	5	4	3	2	1	0	2,047	240
	2	4	x	x	x	x	x	x	9	8	7	6	5	4	3	2	1	0	1,023	480
	3	8	x	x	x	x	x	x	x	8	7	6	5	4	3	2	1	0	511	960
	4	16	x	x	x	x	x	x	x	x	7	6	5	4	3	2	1	0	255	1,920
	5	32	x	x	x	x	x	x	x	x	x	6	5	4	3	2	1	0	127	3,840
1 (14-bit)	0	1	x	x	13	12	11	10	9	8	7	6	5	4	3	2	1	0	16,383	120
	1	2	x	x	x	12	11	10	9	8	7	6	5	4	3	2	1	0	8,191	240
	2	4	x	x	x	x	11	10	9	8	7	6	5	4	3	2	1	0	4,095	480
	3	8	x	x	x	x	x	10	9	8	7	6	5	4	3	2	1	0	2,047	960
	4	16	x	x	x	x	x	x	9	8	7	6	5	4	3	2	1	0	1,023	1,920
	5	32	x	x	x	x	x	x	x	8	7	6	5	4	3	2	1	0	511	3,840

Table 6. PWM Configuration

Functional Description

Local dimming

The Local dimming sequences according to the PWM configuration are shown as follows below, and has two dimming out directions, Head Shift Lighting and Tail Shift Lighting. The set of PWM frequency (pwmclk) according to the Frame frequency can be shown in the Table 7.

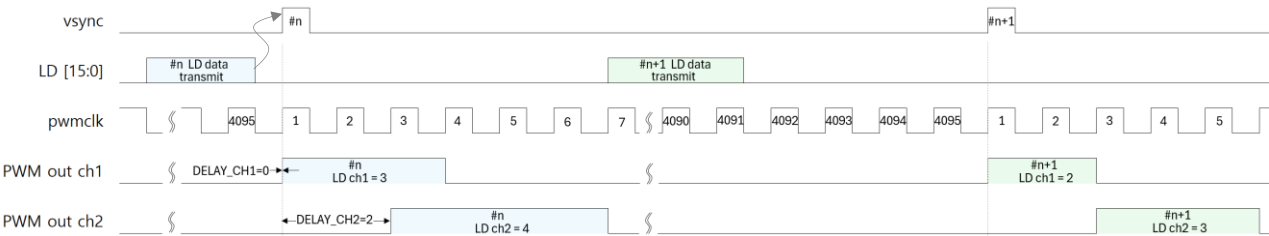


Figure. 6 Head Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=0

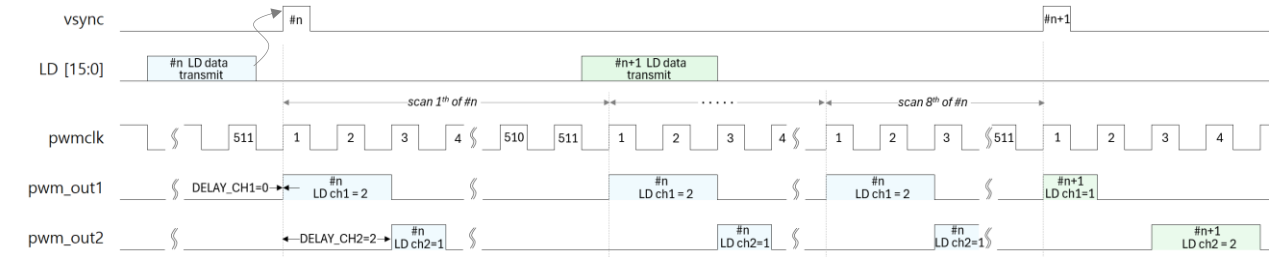


Figure. 7 Head Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=3

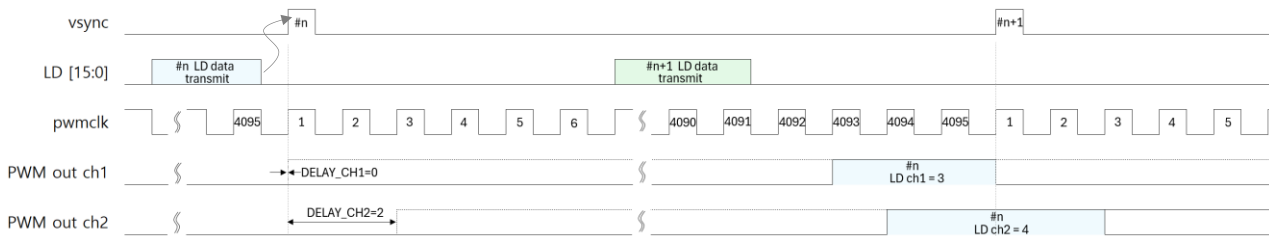


Figure. 8 Tail Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=0

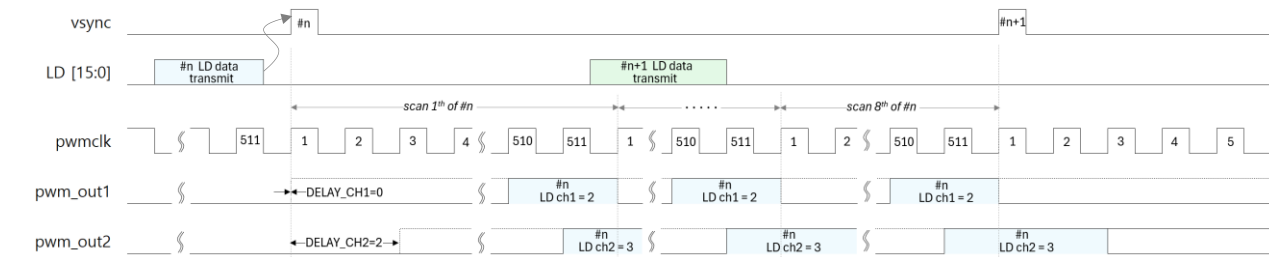


Figure. 9 Tail Shift Lighting @ PWM_RES=0, SCAN_NO[2:0]=3

Functional Description

Optional an external VSYNC and FB

The XD has two external signal modes, VSYNC and FB, and if each external mode is enabled, each function is operated using its external signal instead of using the information obtained from the serial communication. These functions are helpful for the single PCB layout and the connection scheme is shown as follows.

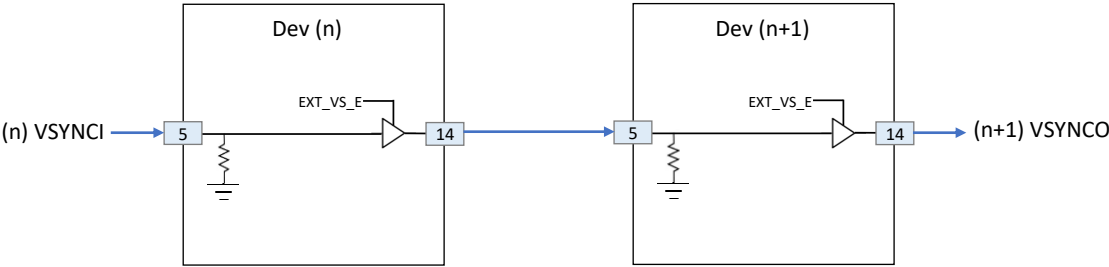


Figure 10. External VSYNC In and Out Scheme

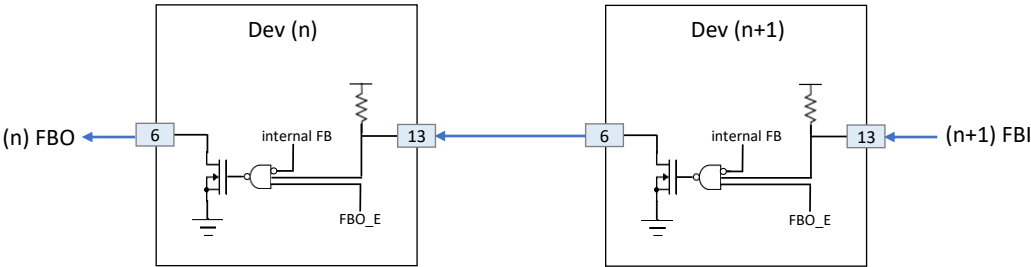


Figure 11. FBI and FBO Scheme

Register Map

RESET / ID

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x00	0x000	RST	x	x	x	x	x	x	ID [4:0]				

Bits	Fields		Descriptions
0	RST	0	NOP
		1	All internal logics are initialized to a default state and this bit will be auto-cleared after internal initialization.
4:0	ID		These bits are read only. If the XD12 will receive the ID-Gen command from the XC24 controller, then the ID value is automatically generated. The ID value is needed for proper reading, so, the ID-Gen command must be firstly executed by the XC24 controller before executing any commands. If all XD12 drivers daisied will receive the ID-Gen command, the last-located device will have the first number of ID, and the first located device will have the last number of ID, that is, the number of daisied devices. The ID value can not be written by a Host.

LD CONTROL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x01	0x000	CH_SIZE [3:0]				EXT_VS_E	EXT_VS_POL	SCAN_NO [2:0]			OVER_TO_E	PWM_RES	LD_DIR

Bits	Fields	Descriptions
		This register determines the dimming sequence. Even if the LD_DIR, PWM_RES, OVER_TO_E and SCAN_NO value will be changed in the middle of the current frame, the value will affect the dimming sequence from the next frame after completing the current frame process.
11:8	CH_SIZE	This value determines how many LD data will be transmitted for one time. This value is not needed to match with the number of enabled channels in the CHANNEL ENABLE register. The LD data to be transmitted will be mapped sequentially from channel 1 to channel n regardless of the corresponded CHx_EN bit activation.
7	EXT_VS_E	This bit determines whether the device activates external VSYNC input or internal VSYNC as the frame sync signal. 0 Internal VSYNC (The internal VSYNC can be generated from the SYNC Command) 1 External VSYNC (The TEST_EN bit should be not activated)
6	EXT_VS_POL	This bit determines the polarity of the external VSYNC input. 0 High active 1 Low active
5:3	SCAN_NO	This value determines the number of scanning within a frame. The scanning number is one if the value is zero. In case the value is greater than 0, the larger the value, the smaller the PWM depth. The available value is from 0 to 5, and the relationship between the set value vs the scanning number and the PWM depth can be showed in below Table 5.
2	OVER_TO_E	Over Turn-On Enable. This bit is an optional and recommended to decide the set according to the dimming performance from internal PWM clock. It is useful in case the PWM clock is faster than ideal frequency under full PWM data.
1	PWM_RES	This bit determines the PWM resolution. 0 12-bit PWM resolution (max steps 4,095) 1 14-bit PWM resolution (max steps 16,383)
0	LD_DIR	This bit determines the direction of modulated edge with increasing on time from a vsync. 0 Head Shift Lighting 1 Tail Shift Lighting

FPWM DIVIDER

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x02	0x000	x	x	x	x	x	FPWM_DIV [6:0]						

Bits	Fields	Descriptions
6:0	FPWM_DIV	This register determines the PWM reference clock. The PWM reference clock can be created by dividing the MCLK frequency by the setting value + 1. Even if the setting value will be changed in the middle of the frame, the value will be applied immediately from the next vsync assertion. The setting value according to the frame frequency is guided as shown Table 7 below.

PWM_RES	Register	100Hz	120Hz	144Hz	165Hz	240Hz
0	FPWM_DIV	95	79	67	59	39
	MCLK_LOCK_CNT	393,120	327,600	278,460	245,700	163,800
1	FPWM_DIV	23	19	16	14	9
	MCLK_LOCK_CNT	393,192	327,660	278,511	245,745	163,830

Table 7. FPWM Configuration

For switching the frame frequency with flicker-less, it is recommended to follow as below sequence.

- 1. Disable the MCLK_LOCK_CNT_E.
- 2. Increase or decrease the FPWM_DIV per frame one by one.
- 3. Change the MCLK_LOCK_CNT suitable to the changed frame frequency.
- 4. Enable the MCLK_LOCK_CNT_E.

CHANNEL ENABLE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x03	0x000	CH12_EN	CH11_EN	CH10_EN	CH9_EN	CH8_EN	CH7_EN	CH6_EN	CH5_EN	CH4_EN	CH3_EN	CH2_EN	CH1_EN

Bits	Fields	Descriptions
11:0	CHx_EN	The CHx_EN determines which channel will be activated. The set value is independent of the value of the CH_SIZE. If the CHx_EN is not set, then the corresponding channel sink output including the its faults, OPENx, SHORTx, FBx, will be deactivated. The CHx_EN bits <i>should be set consecutively</i> , such as CH1_EN, CH2_EN, CH3_EN, and so on, and the number of enabled channels does not need to match the CH_SIZE value.

FAULT STATUS

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
RO	0x05	0x000	MISS_VS	x	x	x	x	x	x	x	THERMAL	SHORT	OPEN	FB

Bits	Fields	Descriptions
11	MISS_VS	The MISS_VS bit indicates whether the vsync missing event happens, and auto-cleared after readout this register.
3	THERMAL	The THERMAL bit indicates whether the THERMAL event due to overheating happens, and this bit will keep asserted as long as the THERMAL event is not cleared. The event threshold can be set by the OFS_TEMP in the TEMP register. If the T_OFF_E bit of the FAULT CONTROL register is set, all output sink drivers will be turned-off as long as the event is asserted, and turned-on again after internal temperature goes down below the set level of the OFS_TEMP.
2	SHORT	The SHORT bit indicates whether the SHORT event happens. The event can be detected in case some channel's headroom goes above the set value in the SHORT_LEVEL of FAULT LEVEL register and its channel's PWM out is active. If the S_OFF_E bit of the FAULT CONTROL register is set, the output of the channels in which the event occurs is turned off. This bit is auto-cleared after readout but set again as long as the event is not cleared.
1	OPEN	The OPEN bit indicates whether the OPEN event happens. The event can be detected in case some channel's headroom goes below 0.25V and its channel's PWM out is inactive. If the O_OFF_E bit of the FAULT CONTROL register is set, the output of the channels in which the event occurs is turned off. This bit is auto-cleared after readout but set again as long as the event is not cleared.
0	FB	The FB bit indicates whether the output channels' headroom voltage is below the value set by FB_LEVEL of the FAULT LEVEL register due to insufficient VLED. This bit is set whenever the FB event occurs in even one of the activated channels and will be auto-cleared after readout this register. This bit will be set again after readout as long as the FB condition is still valid. If the O_FB_E of FAULT CONTROL register is set, the FB event will happen in case the OPEN event is not asserted.

FAULT LEVEL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x06	0x000	x	DEV_MAX_CURR_LEVEL [2:0]			x	SHORT_LEVEL [2:0]			x	FB_LEVEL [2:0]		

Bits	Fields	Descriptions
10:8	DEV_MAX_CURR_LEVEL	The DEV_MAX_CURR_LEVEL determines the physical maximum current level of the device. The max dimming sink-current set by the MAX_CURR_VREF can not be beyond the DEV_MAX_CURR_LEVEL. 3b'000 : max 4mA 3b'001 : max 8mA 3b'010 : max 12mA 3b'011 : max 16mA 3b'100 : max 24mA 3b'101 : max 32mA 3b'110 : max 46mA 3b'111 : max 64mA
6:4	SHORT_LEVEL	The SHORT_LEVEL determines the level at which a SHORT event occurs. 3b'000 : > 3V 3b'001 : > 4V 3b'010 : > 6V 3b'011 : > 8V 3b'100 : > 12V 3b'101 : > 16V 3b'110 : > 24V 3b'111 : > 36V
2:0	FB_LEVEL	The FB_LEVEL determines the level at which a FB event occurs. 3b'000 : < 0.40V 3b'001 : < 0.50V 3b'010 : < 0.60V 3b'011 : < 0.70V 3b'100 : < 0.85V 3b'101 : < 1.00V 3b'110 : < 1.15V 3b'111 : < 1.30V

FAULT CONTROL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x07	0x800	TIMEOUT_E	FBO_E	x	MS_VS_LOCK	MS_VS_DIMM	MS_VS_DET_E	O_FB_E	O_DET_E	S_DET_E	T_OFF_E	S_OFF_E	O_OFF_E

Bits	Fields	Descriptions
11	TIMEOUT_E	If this bit is set, the internal timeout timer starts counting from receiving some read-back command and stops after receiving the first response bit from daisied devices. If the timer will be expired due to any no response, internal communication logics will be recovered to the standby state. The timeout counter expires at approximately 800us upon receiving the read command. Recommended to be set.
10	FBO_E	If this bit is set, the FB In/Out function is enabled. The internal FB status will be driven out active-low directly to the FBO open-drain pin #6 after being ORed with the FBI input pin #13 which is active-low with internally pull-up resistor. This function makes the FB status possible to be supplied in real-time unlike the FB value in the FAULT STATUS register.
9	MS_VS_LOCK	Under the MS_VS_LOCK = 1 and the MS_VS_DET_E = 1, if the vsync missing event happens, then the MCLK_LOCK_CNT function will be halted until receiving a vsync.
8	MS_VS_DIMM	Under the MS_VS_DIMM = 1 and the MS_VS_DET_E = 1, if the vsync missing event happens, then the dimming out will be automatically fixed to the middle level brightness until receiving a vsync.
7	MS_VS_DET_E	This bit enables to detect whether the vsync input is missed or not. If the vsync is not supplied within about 26ms, then the vsync missing event is asserted internally. This event assertion can be readout through the FALUT STATUS register.
5	O_FB_E	This bit setting makes the FB possible to be asserted in only case the corresponding channel's OPEN event is not happen.
4	O_DET_E	This bit setting enables to detect the OPEN event, and the event can be detected in only case the corresponding channel's PWM output is active and its channel's headroom is below 0.25V.
3	S_DET_E	This bit setting enables to detect the SHORT event, and the event can be detected in only case the corresponding channel's PWM output is inactive and its channel's headroom is above the set value in the SHORT_LEVEL register.
2	T_OFF_E	This bit setting makes all channels' output be off in case the THERMAL event happens, and all channels will be activated again after the THERMAL event is cleared.
1	S_OFF_E	This bit setting makes the SHORTed channel's output be off in case the SHORT event happens.
0	O_OFF_E	This bit setting makes the OPENed channel's output be off in case the OPEN event happens.

MAX CURRENT VREF

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x08	0x000	MAX_CURR_VREF [11:0]											

Bits	Fields	Descriptions
11:0	MAX_CURR_VREF	The MAX_CURR_VREF determines the maximum current value of a frame to be dimmed. If the max current level is set to 8mA, then the MAX_CURR_VREF value is proportional to the max current level, and so this value decides the max current reference of all blocks being dimmed for a frame. The setting value will come into effect at the next vsync assertion even if the value is set during current dimming frame earlier than the next vsync.

DELAY CH EXTEND

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x09	0x000	DELAY_CH6 [13:12]		DELAY_CH5 [13:12]		DELAY_CH4 [13:12]		DELAY_CH3 [13:12]		DELAY_CH2 [13:12]		DELAY_CH1 [13:12]	

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0A	0x000	DELAY_CH12 [13:12]		DELAY_CH11 [13:12]		DELAY_CH10 [13:12]		DELAY_CH9 [13:12]		DELAY_CH8 [13:12]		DELAY_CH7 [13:12]	

Bits	Fields	Descriptions
11:0	DELAY_CHx [13:12]	This register includes the MSB 2 bits of each channel's DELAY_CHx [13:0] value. The DELAY_CHx determines from which PWM position the dimming out starts. If this is set to n, then the dimming out starts at (n+1) th PWM position. Normally, the setting value can not exceed the max PWM resolution.

DELAY CH1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0B	0x000	DELAY_CH1 [11:0]											

DELAY CH2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0C	0x000	DELAY_CH2 [11:0]											

DELAY CH3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0D	0x000	DELAY_CH3 [11:0]											

DELAY CH4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0E	0x000	DELAY_CH4 [11:0]											

DELAY CH5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0F	0x000	DELAY_CH5 [11:0]											

DELAY CH6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x10	0x000	DELAY_CH6 [11:0]											

DELAY CH7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x11	0x000	DELAY_CH7 [11:0]											

DELAY CH8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x12	0x000	DELAY_CH8 [11:0]											

DELAY CH9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x13	0x000	DELAY_CH9 [11:0]											

DELAY CH10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x14	0x000	DELAY_CH10 [11:0]											

DELAY CH11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x15	0x000	DELAY_CH11 [11:0]											

DELAY CH12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x16	0x000	DELAY_CH12 [11:0]											

SERIAL CLOCK GEN

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x25	0x318	SERIAL_CLK_LOW [5:0]						SERIAL_CLK_HIGH [5:0]					

Bits	Fields	Descriptions
11:6	SERIAL_CLK_LOW	These bits determine the number of MCLK of V_{DD} level in logic low value '0' or the number of MCLK of zero level in logic high value '1' when the XD12 is driving the D_{IO} pin on read-back. For more details, refer to Table. 4, Logic low time.
5:0	SERIAL_CLK_HIGH	These bits determine the number of MCLK of V_{DD} level in logic high value '1' or the number of MCLK of zero level in logic low value '0' when the XD12 is driving the D_{IO} pin on read-back. For more details, refer to Table. 4, Logic high time.

SERIAL LATENCY

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x26	0x050	x	x	x	x	x	SERIAL_LATENCY [6:0]						

Bits	Fields	Descriptions
6:0	SERIAL_LAT	This determines the interval time between the last bit time of the current command and the first bit time of the next command. The setting value should be larger than the 1-bit serial data time. If not so, the serial data value will be broken.

MCLK LOCK

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x27	0xFB0	MCLK_LOCK_CNT [11:0]											
Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x28	0x84F	MCLK_LOCK_CNT_E	x	x	x	MCLK_LOCK_CNT [19:12]							

Bits	Fields	Descriptions
19:0	MCLK_LOCK_CNT	This determines the FLL target frequency value. The proper value according to the frame frequency is showed in the Table 6.
11	MCLK_LOCK_CNT_E	This bit enables the FLL function. The FLL continues to calibrate the MCLK frequency to exactly match the PWM frequency with the time of one frame. 0 Disable FLL function 1 Enable FLL function

TEMP

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x29	0x800	OFS_TEMP [3:0]				x	x	x	x	x	x	FLT_GAIN [1:0]	

Bits	Fields	Descriptions
11:8	OFS_TEMP	It is recommended to use the default value.
1:0	FLT_GAIN	

OSC FLL MANUAL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x2A	0x000	OSC_FLL_MAN [11:0]											
Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x2B	0x008	OSC_MAN_E								OSC_FLL_MAN [15:12]			

Bits	Fields	Descriptions
11	OSC_MAN_E	Enable the manual mode of the FLL module.
15:0	OSC_FLL_MAN	This determines the FLL target frequency value in manual mode.

These registers can not go outside the Global Technologies.

OP MODE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3F	0x000	TEST_EN	x	x	x	x	PWM_FULL_O	MCLK_O	VREF_O	x	x	x	ADDR_EXT

Bits	Fields	Descriptions
11	TEST_EN	0 Normal operation 1 It is used to test the device for trimming. To use the PWM_FULL_O, MCLK_O and VREF_O function, this bit should be set firstly and cleared lastly after these functions are used.
6	PWM_FULL_O	0 Normal operation 1 Under TEST_EN =1, the enabled channels' PWM out keeps active state regardless of the vsync.
5	MCLK_O	0 Normal operation 1 Under TEST_EN =1, the clock obtained by dividing MCLK by 32 is output to DDIO pin.
4	VREF_O	0 Normal operation 1 Under TEST_EN =1, the MAX_CURR_VREF register value is continuously output to all internal current control logics regardless of the vsync.
0	ADDR_EXT	This bit is used to select which register groups, the general registers or the OTP mirror registers, will be accessed. The OTP control registers address, 0x3A ~ 0x3F, can be commonly accessed regardless of this bit. 0 The general registers can be accessed except for the OTP mirror registers. 1 The OTP mirror registers can be accessed except for the general registers.

OTP ACCESS

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3A	0x000	x	x	x	x	x	x	x	x	OTP_PG_ACCESS_CYCLE [15:12]			

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3B	0x3FF	OTP_PG_ACCESS_CYCLE [11:0]											

OTP WRITE

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3C	0x04	x	x	x	x	x	x	x	x	OTP_WSEL [3:0]			

OTP RD/PROG

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3D	0x000	x	x	x	x	x	x	x	x	x	x	OTP_RD_START	OTP_PG_START

OTP PROTECT

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x3E	0x5A5	PROTECT_EN/DIS [11:0] ※ enable (0x5A5), disable (0xA5A), need to disable for writing											

These registers can not go outside the Global Technologies.

These registers are the OTP Mirror registers.

OTP CRC

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x00	0x000	x	x	x	x	OTP_CRC_CHECKSUM [7:0]							

OSC

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x01	0x020	x	x	x	x	x	x	OSC [5:0]					

VREF CTL

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x02	0x20	x	x	x	x	x	x	VREF_CTL [5:0]					

These registers can not go outside the Global Technologies.

ICTL_L_CH 1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x03	0x40	x	x	x	x	x	ICTL_L_CH1 [6:0]						

ICTL_L_CH 2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x04	0x40	x	x	x	x	x	ICTL_L_CH2 [6:0]						

ICTL_L_CH 3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x05	0x40	x	x	x	x	x	ICTL_L_CH3 [6:0]						

ICTL_L_CH 4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x06	0x40	x	x	x	x	x	ICTL_L_CH4 [6:0]						

ICTL_L_CH 5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x07	0x40	x	x	x	x	x	ICTL_L_CH5 [6:0]						

ICTL_L_CH 6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x08	0x40	x	x	x	x	x	ICTL_L_CH6 [6:0]						

ICTL_L_CH 7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x09	0x40	x	x	x	x	x	ICTL_L_CH7 [6:0]						

ICTL_L_CH 8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0A	0x40	x	x	x	x	x	ICTL_L_CH8 [6:0]						

ICTL_L_CH 9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0B	0x40	x	x	x	x	x	ICTL_L_CH9 [6:0]						

ICTL_L_CH 10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0C	0x40	x	x	x	x	x	ICTL_L_CH10 [6:0]						

ICTL_L_CH 11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0D	0x40	x	x	x	x	x	ICTL_L_CH11 [6:0]						

ICTL_L_CH 12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x0E	0x40	x	x	x	x	x	ICTL_L_CH12 [6:0]						

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ICTL_H_CH 1

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1B	0x40	x	x	x	x	x	ICTL_H_CH1 [6:0]						

ICTL_H_CH 2

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1C	0x40	x	x	x	x	x	ICTL_H_CH2 [6:0]						

ICTL_H_CH 3

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1D	0x40	x	x	x	x	x	ICTL_H_CH3 [6:0]						

ICTL_H_CH 4

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1E	0x40	x	x	x	x	x	ICTL_H_CH4 [6:0]						

ICTL_H_CH 5

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x1F	0x40	x	x	x	x	x	ICTL_H_CH5 [6:0]						

ICTL_H_CH 6

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x20	0x40	x	x	x	x	x	ICTL_H_CH6 [6:0]						

ICTL_H_CH 7

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x21	0x40	x	x	x	x	x	ICTL_H_CH7 [6:0]						

ICTL_H_CH 8

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x22	0x40	x	x	x	x	x	ICTL_H_CH8 [6:0]						

ICTL_H_CH 9

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x23	0x40	x	x	x	x	x	ICTL_H_CH9 [6:0]						

ICTL_H_CH 10

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x24	0x40	x	x	x	x	x	ICTL_H_CH10 [6:0]						

ICTL_H_CH 11

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x25	0x40	x	x	x	x	x	ICTL_H_CH11 [6:0]						

ICTL_H_CH 12

Acc	Addr	Default	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0
R/W	0x26	0x40	x	x	x	x	x	ICTL_H_CH12 [6:0]						

Soldering Temperature Reflow Profile

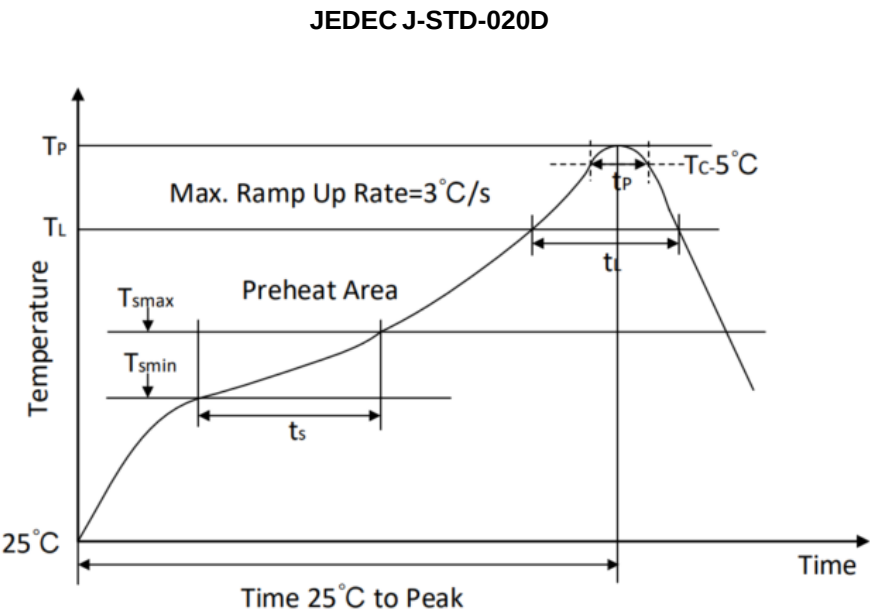


Fig 9. Reflow Soldering information

Table 5. Reflow Soldering Parameter

Parameter	Pb-free soldering conditions
Average ramp-up rate (217 °C to Peak)	3°C/second max.
Time of Preheat temp (from 150 °C to 200 °C)	60 ~ 120 second
Time to be maintained above 217 °C	60 ~ 150 second
Peak temperature	260 +5/-0 °C
Time within 5 °C of actual peak temp	30 second
Ramp-down rate	6 °C/second max.
Time from 25°C to peak temp	8 minutes max.

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

Package specification

QFN 3030-24L Package Outline

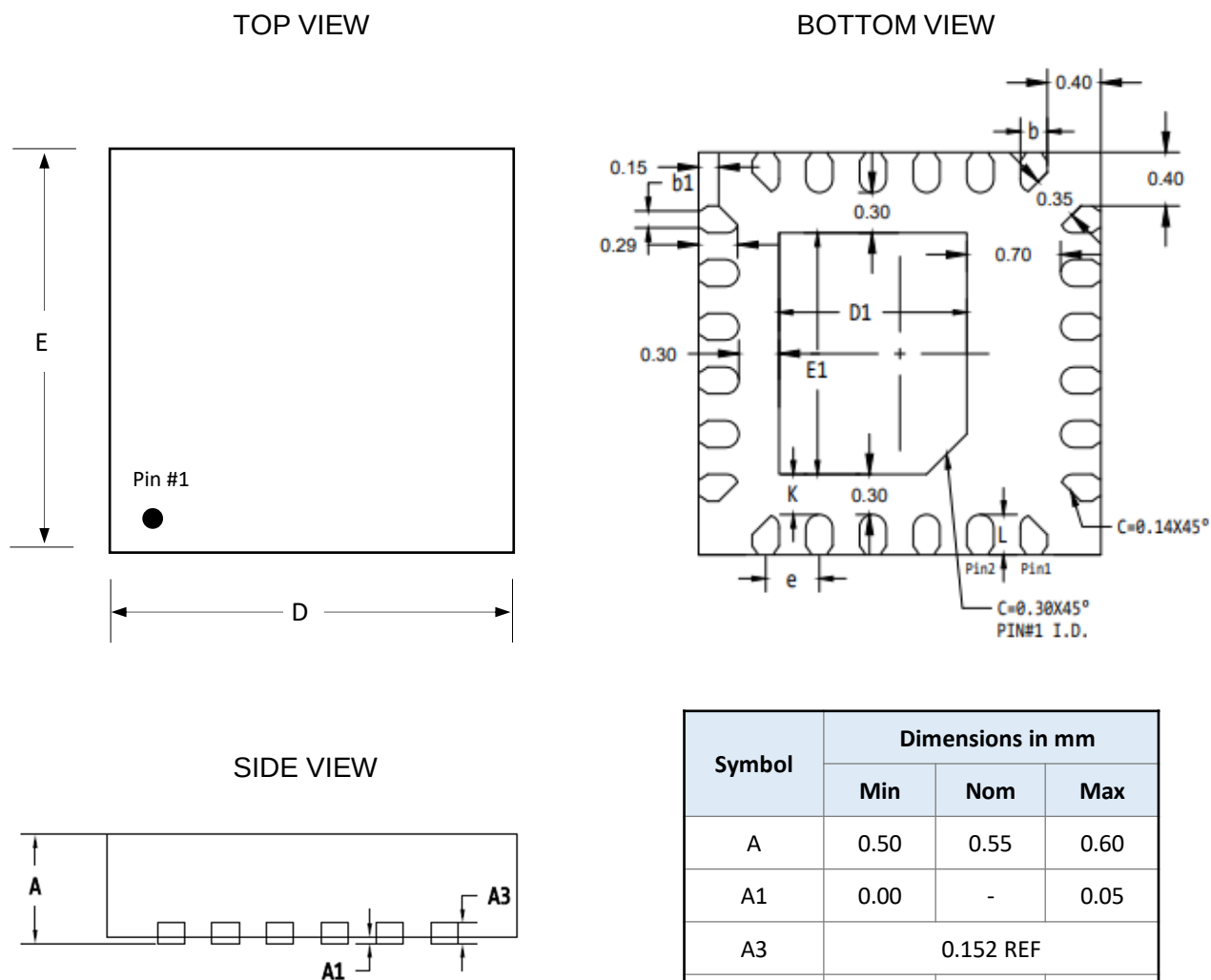


Fig 10. Package Outline

Symbol	Dimensions in mm		
	Min	Nom	Max
A	0.50	0.55	0.60
A1	0.00	-	0.05
A3	0.152 REF		
b	0.15	0.20	0.25
b1	0.08	0.13	0.18
D	2.90	3.00	3.10
D1	1.25	1.30	1.35
E	2.90	3.00	3.10
E1	1.75	1.80	1.85
e	0.40 BSC		
K	0.25	0.30	0.35
L	0.25	0.30	0.35

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Recommended Footprint

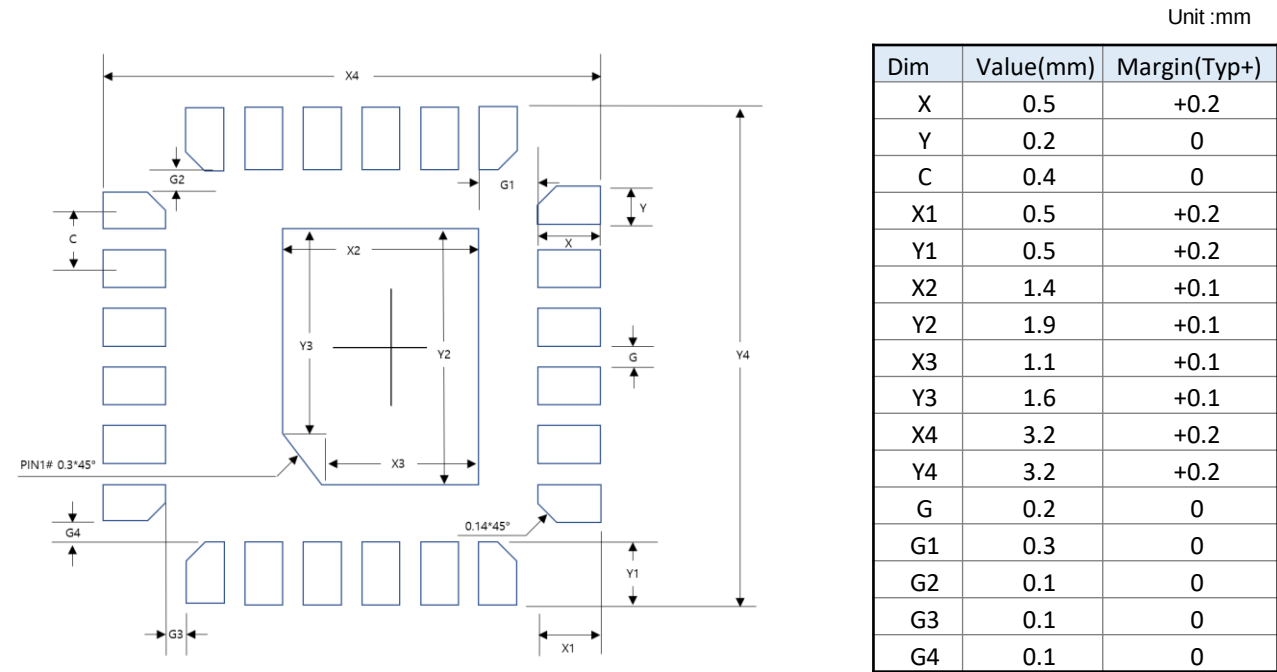


Fig 11. Recommended Footprint

Note: The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD- 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”

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